HEIGHTS ON ABELIAN GROUPS, WITH SOLUTIONS

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The Mordell–Weil theorem has two steps. The weak Mordell–Weil theorem says that $E(K)/mE(K)$ is finite, and a height function turns this into finite generation [Sil09, §VIII.3]. The second step does not use the curve: it works for any abelian group with a height that is quadratic up to a bounded error. This worksheet is about such heights on an abstract abelian group: which functions are heights of which kind, what their $O(1)$ constants are, and how a quadratic height behaves. Problem 1.2 sets up the definitions and the first estimates. Problem 2.2, the heart of the worksheet, proves the descent theorem and shows, on $\mathbb{Q}$ and on a direct sum of copies of $\mathbb{Z}$, that its hypotheses are needed. Problem 3.2 is the algebra of quadratic heights. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An abelian group A, written additively, with elements P, Q, R, torsion subgroup A_{tors}, and, for an integer $m \geqslant 2$, the quotient $A/m \coloneqq A/mA$.
- For real functions f, g on a set X, the notation $f = g + O(1)$, meaning that $f - g$ is bounded on X. An inequality $f \leqslant g + c$ that holds for one constant c holds for every larger one, so the constants that bound heights are taken non-negative, and a lower bound is written $-c$ with $c \geqslant 0$.

How to use this. Allow about 20 minutes for Problem 1.2, 20 for Problem 2.2 and 20 for Problem 3.2. Problem 2.2 is the one not to skip. Everything is by hand.

1. Heights and the parallelogram law

This section is the vocabulary of [Sil09, §VIII.3], and the estimates [Sil09, Corollary VIII.6.4] that the heights on an elliptic curve satisfy, proved here for any height that satisfies the parallelogram law up to a bounded error [HS00, §B.4].

Definition 1.1 (Heights [Sil09, §§VIII.3, VIII.6])**.** A height on a set X is a function $h\colon X \to \mathbb{R}$. Two heights h, h' on X are equivalent, written $h = h' + O(1)$, if $h - h'$ is bounded on X. A height h has the Northcott property if $\{x \in X : h(x) \leqslant C\}$ is finite for every $C \in \mathbb{R}$. The defect of a height h on A is the function

$$\delta_h(P, Q) \coloneqq h(P + Q) + h(P - Q) - 2h(P) - 2h(Q)$$

on $A \times A$. The height h is quasi-quadratic if δ_h is bounded, and quadratic if $\delta_h = 0$ (the parallelogram law).

Problem 1.2 (Heights and the parallelogram law)**.**
(1) Show that equivalence of heights on a set X is an equivalence relation. Show that the Northcott property depends only on the equivalence class, and that a height with the Northcott property is bounded below. Give a height on $\mathbb{Z}$ that is bounded below and does not have the Northcott property.

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Solution. **If $|h - h'| \leqslant c$, then $\{h' \leqslant C\} \subseteq \{h \leqslant C + c\}$.** The difference $h - h$ is zero, $h' - h$ is bounded when $h - h'$ is, and if $|h - h'| \leqslant c$ and $|h' - h''| \leqslant c'$, then $|h - h''| \leqslant c + c'$. If $|h - h'| \leqslant c$ and $h'(x) \leqslant C$, then $h(x) \leqslant C + c$, which gives the inclusion of the answer. So if h has the Northcott property, the set $\{h' \leqslant C\}$ is contained in a finite set. For the lower bound, let $x_0 \in X$. The set $\{h \leqslant h(x_0)\}$ is finite and contains x_0, so it has a minimum, and that minimum is a lower bound for h on X. The zero height on $\mathbb{Z}$ is bounded below, and $\{h \leqslant 0\} = \mathbb{Z}$. A more interesting example is the number $\omega(n)$ of distinct prime divisors of $n \neq 0$, with $\omega(0) \coloneqq 0$. It is non-negative, and the set $\{\omega \leqslant 1\}$ contains every prime power, so it is infinite.

(2) Let h be a quasi-quadratic height on A, and let $c \geqslant 0$ be a constant with $|\delta_h| \leqslant c$. Show that $|h(0)| \leqslant c/2$, that $|h(-P) - h(P)| \leqslant 2c$, and that $|h(2P) - 4h(P)| \leqslant 3c/2$ for every $P \in A$.

Solution. By definition, $\delta_h(0,0) = -2h(0)$, $\delta_h(0,P) = h(-P) - h(P) - 2h(0)$ and $\delta_h(P,P) = h(2P) - 4h(P) + h(0)$, using $h(P-0) = h(P)$ and $h(P-P) = h(0)$. The first gives $|h(0)| \leqslant c/2$. The second gives $|h(-P) - h(P)| \leqslant c + 2|h(0)| \leqslant 2c$, and the third gives $|h(2P) - 4h(P)| \leqslant c + |h(0)| \leqslant 3c/2$.

(3) Let h be a quasi-quadratic height on A, and let $c \geqslant 0$ be a constant with $|\delta_h| \leqslant c$. Show that for every $n \in \mathbb{Z}$ there is a constant $d_n \geqslant 0$ with $|h(nP) - n^2 h(P)| \leqslant d_n$ for all $P \in A$, and give a recursion for d_n in terms of c. Suppose moreover that $h \geqslant -c'$ for a constant $c' \geqslant 0$. Show that for every $Q \in A$ there is a constant $C(Q) \geqslant 0$ with $h(P + Q) \leqslant 2h(P) + C(Q)$ for all $P \in A$, and give $C(Q)$ explicitly.

Solution. **$d_{n+1} = 2d_n + d_{n-1} + c$, $d_0 = c/2$, $d_1 = 0$, $C(Q) = 2h(Q) + 2c' + c$.** Fix $P \in A$, and let $e_n \coloneqq h(nP) - n^2 h(P)$ for $n \in \mathbb{Z}$. By (2), $|e_0| = |h(0)| \leqslant c/2$, and $e_1 = 0$. So $d_0 = c/2$ and $d_1 = 0$ work.
Let $n \geqslant 1$. Since $(n+1)^2 + (n-1)^2 = 2n^2 + 2$, the defect at (nP, P) is

$$\delta_h(nP, P) = h((n+1)P) + h((n-1)P) - 2h(nP) - 2h(P) = e_{n+1} + e_{n-1} - 2e_n.$$

So $e_{n+1} = 2e_n - e_{n-1} + \delta_h(nP, P)$, and $|e_{n+1}| \leqslant 2d_n + d_{n-1} + c$. By induction on n, the recursion of the answer gives a constant d_n for every $n \geqslant 0$. For negative multiples, (2) applied to nP gives $|h(-nP) - h(nP)| \leqslant 2c$, so $|e_{-n}| \leqslant |e_n| + 2c$, and $d_{-n} \coloneqq d_n + 2c$ works. None of these bounds depends on P.
For the last claim, the defect at (P, Q) gives

$$h(P + Q) = 2h(P) + 2h(Q) - h(P - Q) + \delta_h(P, Q) \leqslant 2h(P) + 2h(Q) + c' + c,$$

since $h(P - Q) \geqslant -c'$ and $\delta_h \leqslant c$. The constant $2h(Q) + c' + c$ can be negative. Enlarging it to $C(Q) \coloneqq 2h(Q) + 2c' + c$ makes it at least $c \geqslant 0$, since $h(Q) \geqslant -c'$.

Remark. The recursion gives $d_2 = 3c/2$, $d_3 = 4c$ and $d_4 = 21c/2$, and in general d_n grows like $(1 + \sqrt{2})^n$. A bound of the order of n^2 is true, and comes with the canonical height, in a later worksheet.

(4) Let $f \colon A \to \mathbb{R}$ be a homomorphism. Show that the height f^2 on A is quadratic. Show that a sum of quadratic heights on A is quadratic.

Solution. **$\delta_{f^2} = 0$, and $\delta_{h+h'} = \delta_h + \delta_{h'}$.** Let $a \coloneqq f(P)$ and $b \coloneqq f(Q)$. Then $f(P \pm Q) = a \pm b$, and

$$\delta_{f^2}(P, Q) = (a + b)^2 + (a - b)^2 - 2a^2 - 2b^2 = 0,$$

which is the parallelogram law in $\mathbb{R}$. Each term of the defect is linear in the height, so $\delta_{h+h'} = \delta_h + \delta_{h'}$, and a sum of heights with zero defect has zero defect.

(5) Let $A = \mathbb{Z}^2$, $q(x, y) \coloneqq x^2 + y^2$ and $q'(x, y) \coloneqq x^2 + (x + y)^2$. Show that q and q' are quadratic heights with the Northcott property, that $q \leqslant 3q'$ and $q' \leqslant 3q$, and that they are not equivalent.

SOLUTION. **$q' - q = x(x+2y)$ is unbounded.** By (4), applied to the homomorphisms x, y and $x+y$, both q and q' are quadratic. If $q \leqslant C$ then $|x|, |y| \leqslant \sqrt{C}$, so q has the Northcott property. From $(x+y)^2 \leqslant 2x^2 + 2y^2$ we get $q' \leqslant 3x^2 + 2y^2 \leqslant 3q$, and from $y^2 = ((x+y) - x)^2 \leqslant 2(x+y)^2 + 2x^2$ we get $q \leqslant 3q'$. So q' has the Northcott property as well. But $q'(n,n) - q(n,n) = 3n^2$ is unbounded.

Remark. Two heights with $q \leqslant 3q'$ and $q' \leqslant 3q$ are comparable, which is weaker than equivalent. So a quadratic height is far from determined by its equivalence class, although an equivalence class contains at most one quadratic height, as a later worksheet shows.

2. The descent theorem

This section is [Sil09, Theorem VIII.3.1] and its proof, for an abstract abelian group, and the examples of groups that fail one of its hypotheses. The count of rational numbers of bounded height is [HS00, Theorem B.2.3].

Theorem 2.1 (Descent theorem [Sil09, Theorem VIII.3.1])**.** *Let h be a height on A with the following properties.*

(1) For every $Q \in A$ there is a constant $C_1(Q)$ with $h(P+Q) \leqslant 2h(P) + C_1(Q)$ for all $P \in A$.
(2) There are an integer $m \geqslant 2$ and a constant C_2 with $h(mP) \geqslant m^2 h(P) - C_2$ for all $P \in A$.
(3) The height h has the Northcott property.

If A/m is finite for the m of (2), then A is finitely generated.

Problem 2.2 (The descent theorem)**.**

(1) Let h be a quasi-quadratic height on A with the Northcott property. Show that h satisfies the hypotheses of Theorem 2.1 for every $m \geqslant 2$.

SOLUTION. **$C_1(Q) = C(Q)$ and $C_2 = d_m$, from Problem 1.2(3).** By Problem 1.2(1), $h \geqslant -c'$ for some $c' \geqslant 0$. Then Problem 1.2(3) gives $h(P+Q) \leqslant 2h(P) + C(Q)$, which is (1), and $h(mP) \geqslant m^2 h(P) - d_m$, which is (2), for every $m \geqslant 2$.

(2) Let h be a height on A with the Northcott property, let $m \geqslant 2$ be an integer, and let $0 < a < b$ be real numbers. Suppose that for every $Q \in A$ there is a constant $C_1(Q) \geqslant 0$ with $h(P+Q) \leqslant a\,h(P) + C_1(Q)$ for all P, and that there is a constant $C_2 \geqslant 0$ with $h(mP) \geqslant b\,h(P) - C_2$ for all P. Suppose also that A/m is finite, let $Q_1, \dots, Q_s$ represent its cosets, and let $C := \max_i C_1(-Q_i) + C_2$. Show that A is generated by $Q_1, \dots, Q_s$ and the finitely many $P \in A$ with $h(P) \leqslant 1 + C/(b-a)$.

SOLUTION. **If $P_{j-1} = mP_j + Q_{i_j}$, then $h(P_j) \leqslant \rho\, h(P_{j-1}) + C/b$ with $\rho := a/b < 1$.** Given $P_0 := P$, choose P_j and i_j with $P_{j-1} = mP_j + Q_{i_j}$. Then

$$b\,h(P_j) \leqslant h(mP_j) + C_2 = h(P_{j-1} - Q_{i_j}) + C_2 \leqslant a\,h(P_{j-1}) + C.$$

Since $\rho > 0$, induction gives

$$h(P_n) \leqslant \rho^n h(P) + \frac{C}{b}\bigl(1 + \rho + \cdots + \rho^{n-1}\bigr) \leqslant \rho^n h(P) + \frac{C}{b-a},$$

where the last step uses $C \geqslant 0$. For n large, $\rho^n h(P) \leqslant 1$, so $h(P_n) \leqslant 1 + C/(b-a)$, and $P = m^n P_n + \sum_{j=1}^{n} m^{j-1} Q_{i_j}$. The set $\{h \leqslant 1 + C/(b-a)\}$ is finite by the Northcott property. No lower bound on h is needed.

Remark. Silverman's case is $a = 2$ and $b = m^2 \geqslant 4$, where $C/(b-a) \leqslant C/2$ recovers his bound $1 + \frac{1}{2}(C_1 + C_2)$ [Sil09, proof of Theorem VIII.3.1].

(3) Let $A = \mathbb{Q}$, as an additive group, with the logarithmic Weil height

$$h(u/w) := \log \max(|u|, |w|),$$

for u/w in lowest terms. Show that h has the Northcott property, that $h(x+y) \leqslant h(x)+h(y)+\log 2$, with equality for $x = y = 1$, that $h(mx) \leqslant h(x) + \log m$ and $\mathbb{Q}/m = 0$ for every integer $m \geqslant 2$. Which hypothesis of (2) fails, for every $m \geqslant 2$ and every $b > 1$? Is h quasi-quadratic?

Solution. **The second, since $h(mx) \leqslant h(x)+\log m$. Not quasi-quadratic: $\delta_h(n,n) = \log 2 - 3\log n$.** There are at most $(2e^C+1)^2$ pairs (u,w) of integers with $|u|, |w| \leqslant e^C$, so $\{h \leqslant C\}$ is finite. From $u/w + u'/w' = (uw' + u'w)/(ww')$, before canceling, $\max(|uw' + u'w|, |ww'|) \leqslant 2\max(|u|,|w|)\max(|u'|,|w'|)$, and canceling only lowers the height. For $x = y = 1$ both sides are $\log 2$. So the first hypothesis holds with $a = 1$ and $C_1(Q) = h(Q) + \log 2$, since $h \geqslant 0$. Likewise $h(mu/w) \leqslant \log(m \max(|u|,|w|))$. Every rational number is m times a rational number, so $\mathbb{Q}/m = 0$. But $\mathbb{Q}$ is not finitely generated: a finitely generated subgroup of $\mathbb{Q}$ has a common denominator. So the second hypothesis fails, and directly: if $h(mx) \geqslant b\,h(x) - C_2$ with $b > 1$, then $(b-1)h(x) \leqslant C_2 + \log m$ for every x, contradicting the Northcott property on the infinite set $\mathbb{Q}$. Finally $\delta_h(n,n) = h(2n) + h(0) - 4h(n) = \log(2n) - 4\log n$ for $n \geqslant 1$, which is unbounded.

(4) Let $A = \bigoplus_{i \geqslant 1} \mathbb{Z}$, let $w_1, w_2, \ldots$ be positive real numbers, and $h(x) := \sum_i w_i x_i^2$. Show that h is quadratic. For which weights w_i does h have the Northcott property? For such weights, which hypothesis of Theorem 2.1 fails for A?

Solution. **Northcott if and only if $w_i \to \infty$. The quotient A/m is infinite.** Each map $x \mapsto \sqrt{w_i}\, x_i$ is a homomorphism $A \to \mathbb{R}$, so each $w_i x_i^2$ satisfies the parallelogram law, by Problem 1.2(4), and the sums are finite. Suppose $w_i \to \infty$ and let $C \in \mathbb{R}$. Only finitely many i have $w_i \leqslant C$. If $h(x) \leqslant C$ and $x_i \neq 0$, then $w_i \leqslant w_i x_i^2 \leqslant C$, so x is supported on those i, with $|x_i| \leqslant \sqrt{C/w_i}$. So $\{h \leqslant C\}$ is finite. If w_i does not tend to infinity, then $w_i \leqslant C$ for infinitely many i, and the corresponding basis vectors e_i have $h(e_i) = w_i \leqslant C$. For $w_i = i$, the height h satisfies every hypothesis on h, by (1), but $A/m \cong \bigoplus_{i \geqslant 1} \mathbb{Z}/m\mathbb{Z}$ is infinite, and A is not finitely generated.

Remark. Conversely, let A be finitely generated, and choose an isomorphism $A \cong A_{\text{tors}} \times \mathbb{Z}^r$. The sum of the squares of the r coordinates is a quadratic height on A, and $\{h \leqslant C\}$ is the preimage of finitely many elements of $\mathbb{Z}^r$, each preimage a coset of the finite group A_{tors}. So A is finitely generated if and only if $A/2$ is finite and A has a quasi-quadratic height with the Northcott property.

3. Quadratic heights and their pairing

This section is [HS00, Lemma B.5.2], the algebra of a height that satisfies the parallelogram law exactly, which [Sil09, Theorem VIII.9.3(c)] states for the canonical height on an elliptic curve.

Definition 3.1 (The pairing of a quadratic height [HS00, §B.5])**.** Let q be a quadratic height on A. Its pairing is

$$\langle P, Q \rangle_q := q(P+Q) - q(P) - q(Q).$$

Problem 3.2 (Quadratic heights and their pairing)**.**
Let q be a quadratic height on A.

(1) Show that $q(0) = 0$, $q(-P) = q(P)$ and $q(2P) = 4q(P)$, and that $\langle\ ,\ \rangle_q$ is symmetric and bilinear. Deduce that $\langle P, P \rangle_q = 2q(P)$ and $q(nP) = n^2 q(P)$ for every $n \in \mathbb{Z}$. Conversely, show

that for a symmetric bilinear map $B\colon A\times A\to\mathbb{R}$, the function $P\mapsto\frac12 B(P,P)$ is a quadratic height with pairing B.

SOLUTION. **The four laws combine to $\langle P+Q,R\rangle_q=\langle P,R\rangle_q+\langle Q,R\rangle_q$.** The parallelogram law with $P=Q=0$ gives $2q(0)=4q(0)$, with $P=0$ it gives $q(Q)+q(-Q)=2q(Q)$, and with $Q=P$ it gives $q(2P)=4q(P)$. The pairing is symmetric by definition. The parallelogram law for the pairs $(X,Y)=(P+R,Q)$, $(P,R-Q)$, $(P+Q,R)$ and (R,Q) gives

$$q(P+R+Q)+q(P+R-Q)=2q(P+R)+2q(Q), \tag{3.1}$$

$$q(P+R-Q)+q(P-R+Q)=2q(P)+2q(R-Q), \tag{3.2}$$

$$q(P+Q+R)+q(P+Q-R)=2q(P+Q)+2q(R), \tag{3.3}$$

$$q(R+Q)+q(R-Q)=2q(R)+2q(Q). \tag{3.4}$$

Take

$$(3.1)-(3.2)+(3.3)-2\,(3.4).$$

On the left, the two terms $q(P+R-Q)$ cancel, $q(P-R+Q)$ cancels against $q(P+Q-R)$, and $q(P+R+Q)$ and $q(P+Q+R)$ add up. The term $2q(R-Q)$ appears on both sides. Dividing by 2 gives

$$q(P+Q+R)-q(P+Q)-q(P+R)-q(Q+R)+q(P)+q(Q)+q(R)=0,$$

which says that $\langle P+Q,R\rangle_q=\langle P,R\rangle_q+\langle Q,R\rangle_q$. Then $\langle P,P\rangle_q=q(2P)-2q(P)=2q(P)$, and by bilinearity $q(nP)=\frac12\langle nP,nP\rangle_q=n^2q(P)$. Conversely, for $q(P):=\frac12 B(P,P)$, expanding $B(P\pm Q,P\pm Q)$ gives $q(P+Q)+q(P-Q)=B(P,P)+B(Q,Q)=2q(P)+2q(Q)$ and $q(P+Q)-q(P)-q(Q)=B(P,Q)$.

Remark. So $q\mapsto\langle\ ,\ \rangle_q$ and $B\mapsto\frac12 B(P,P)$ are inverse bijections between quadratic heights on A and symmetric bilinear maps $A\times A\to\mathbb{R}$.

(2) Let $T\in A_{\mathrm{tors}}$. Show that $q(T)=0$, that $\langle P,T\rangle_q=0$ for every $P\in A$, and that $q(P+T)=q(P)$. So q and its pairing factor through A/A_{tors}.

SOLUTION. **$n\langle P,T\rangle_q=\langle P,nT\rangle_q=\langle P,0\rangle_q=0$.** Let $n\geqslant 1$ with $nT=0$. By (1), $n^2q(T)=q(nT)=q(0)=0$, and $\langle P,0\rangle_q=q(P)-q(P)-q(0)=0$, which gives the answer. Then $q(P+T)=q(P)+q(T)+\langle P,T\rangle_q=q(P)$.

(3) Let $A=\mathbb{Z}^r$ with standard basis $e_1,\dots,e_r$. Show that every quadratic height on A is $q(x)=\frac12 x^{\mathsf T}Gx$ for a unique real symmetric $r\times r$ matrix G, and that every symmetric G arises. Show that if G is positive definite, then q has the Northcott property.

SOLUTION. **$G_{ij}=\langle e_i,e_j\rangle_q$, and $q(x)\geqslant\frac12\lambda|x|^2$ for the least eigenvalue λ of G.** By bilinearity, $\langle x,y\rangle_q=\sum_{i,j}x_iy_j\langle e_i,e_j\rangle_q=x^{\mathsf T}Gy$ with $G_{ij}:=\langle e_i,e_j\rangle_q$, which is symmetric, and $q(x)=\frac12\langle x,x\rangle_q$. The matrix is unique because G_{ij} is determined by q. Conversely, $(x,y)\mapsto x^{\mathsf T}Gy$ is symmetric and bilinear for every symmetric G, so (1) applies. If G is positive definite, then $x^{\mathsf T}Gx\geqslant\lambda|x|^2$, so $\{q\leqslant C\}$ is contained in the finite set $\{x\in\mathbb{Z}^r:|x|^2\leqslant 2C/\lambda\}$.

Remark. The converse holds: a quadratic height on $\mathbb{Z}^r$ with the Northcott property is positive definite [Sil09, Lemma VIII.9.5]. Its proof needs Minkowski's theorem, and [Sil09, Exercise 8.16] shows why: $q(x,y)=(x-\sqrt{2}\,y)^2$ is positive on $\mathbb{Z}^2\setminus\{0\}$ without being positive definite. That is for next time. Note that a quadratic height can be negative, as $G=-1$ on $\mathbb{Z}$ shows.

REFERENCES

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